

EXPT NO: 8	OPTICAL FLOW COMPUTATION ALGORITHM
DATE: 18/10/2025	

AIM:

Implement optical flow computation algorithm.

ALGORITHM:

1. Read two **consecutive video frames**.
2. Convert both to **grayscale**.
3. Compute optical flow using **cv2.calcOpticalFlowFarneback()**.
4. Visualize motion vectors using **color coding**.
5. Overlay flow on **original image**.
6. Display **motion visualization**.

CODE:

```
import cv2

import numpy as np
from google.colab.patches import cv2_imshow

cap = cv2.VideoCapture('video.mp4')
ret, old = cap.read()
old_gray = cv2.cvtColor(old, cv2.COLOR_BGR2GRAY)

# Initial feature points
p0 = cv2.goodFeaturesToTrack(old_gray, maxCorners=50, qualityLevel=0.3,
minDistance=7)
mask = np.zeros_like(old)

# Parameters for Lucas-Kanade optical flow
lk = dict(winSize=(15, 15), maxLevel=2,
          criteria=(cv2.TERM_CRITERIA_EPS | cv2.TERM_CRITERIA_COUNT, 10,
0.03))
```

```

frame_count = 0
while True:
    ret, frame = cap.read()
    if not ret:
        break
    frame_gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Calculate optical flow
    p1, st, _ = cv2.calcOpticalFlowPyrLK(old_gray, frame_gray, p0, None, **lk)

    # Select good points
    good_new = p1[st == 1]
    good_old = p0[st == 1]

    # Draw motion vectors every 5th frame only
    if frame_count % 5 == 0:
        mask[:] = 0 # clear mask for clean output
        for (new, old_pt) in zip(good_new, good_old):
            a, b = new.ravel()
            c, d = old_pt.ravel()
            mask = cv2.line(mask, (int(a), int(b)), (int(c), int(d)), (0, 255,
0), 2)
            frame = cv2.circle(frame, (int(a), int(b)), 3, (0, 0, 255), -1)
            motion = np.mean(np.linalg.norm(good_new - good_old, axis=1))
            print(f"Frame {frame_count}: Avg motion = {motion:.2f}")
            cv2_imshow(cv2.add(frame, mask))

    # Update previous frame and points
    old_gray = frame_gray.copy()
    p0 = good_new.reshape(-1, 1, 2)
    frame_count += 1

    if cv2.waitKey(30) & 0xFF == 27: # ESC to quit
        break

cap.release()
cv2.destroyAllWindows()

```

OUTPUT:



RESULT:

Thus, optical flow computation algorithm was implemented successfully.